



ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Consider potential ethical problems, and potential solutions – both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

PROBLEM	An incorrect AI recommendation could delay treatment if accepted without clinical judgement.		SOLUTION
	USER	The triage nurse reviews every recommendation alongside the patient's assessment and overrides the AI when clinically appropriate.	
	ROBOT	The dashboard states AI is decision support only and requires nurse confirmation before a triage level is recorded.	

Data security

Is the robot in a unique data collection position? How is the user's data protected?

PROBLEM	Sensitive patient information could be exposed through unauthorised access or insecure systems.		SOLUTION
	USER	Follow hospital privacy policies, use secure logins, and log out when leaving the workstation.	
	ROBOT	Encrypt all patient data, enforce role-based access, and maintain audit logs for every recommendation and override.	

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

PROBLEM	Clinicians may not trust or understand recommendations if the reasoning is unclear.		SOLUTION
	USER	The nurse reviews supporting clinical information before accepting the recommendation.	
	ROBOT	Display the predicted ESI level, confidence score, and the most influential patient factors.	

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

PROBLEM	AI predictions may perform differently across patient populations if bias exists in the training data.	Report unexpected or inconsistent recommendations through clinical governance processes.	SOLUTION
	USER	ROBOT	
		Regularly validate performance across age, gender and patient groups; retrain using representative data.	

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

PROBLEM	Over-reliance on AI may reduce clinician confidence or discourage independent decision-making.		SOLUTION
	USER	Maintain professional clinical judgement; use AI only as an additional source of information.	
	ROBOT	Provide clear reminders that the nurse has final authority; make overriding quick and straightforward.	

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

PROBLEM	Users may skip parts of the clinical assessment because they trust the AI recommendation.		SOLUTION
	USER	Complete the standard Mercer ED triage assessment before relying on AI outputs.	
	ROBOT	Require mandatory assessment fields before generating recommendations; log overrides for audit.	

